402Pilot: Technical Supplement
Ablation and Appendices

Supplement to the Course Project Report

A Notation and Locked Settings

This appendix is a compact legend for the symbols used by the ablation and appendices in this supplement. Definitions and derivations remain in the course report; Table 1 records the convention and locked value used in the released artifact.

The paper writes a single discount γ because the released experiments set the utility and cost discounts equal. The implementation exposes a separate cost-discount parameter γ_c for future variants, but it is not varied in the reported results.

Table 1: Compact notation legend.

Symbol	Meaning	Domain / locked value
Rounds, arms, and context		
t, T	Round index and horizon	$t \in \{0, \dots, T-1\}$; $T = 10{,}000$
a, K	Provider arm and number of arms	$K = 5$: P-cheap, P-mid, P-premium, P-adv, P-flaky
a_t	Arm selected at round t	$a_t \in \mathcal{A}_t$
$\mathbf{x}_t$	Task context vector	$\mathbf{x}_t \in \mathbb{R}^d$, $d = 7$ in the released encoder
$k(\mathbf{x}_t), K_b$	Context bucket map and number of buckets	T1/T2/T3a/T3b; $K_b = 4$ by default
Payment loop and feedback		
Q_t	Quoted candidate set from discovery/pricing	Provider-price pairs before selection
$p_{a,t}$	Effective charge for arm a at round t	May drift; revealed numerically only for selected calls
$\mathcal{A}_t$	Affordable provider set	$\{a : p_{a,t} \le B_t\}$, exposed to the policy
q_t	Observed quality of the chosen arm	$[0, 1]$
f_t	Failure flag	$\{0, 1\}$ for timeout, parse failure, or non-delivery
c_t	Realized charge	$c_t = p_{a_t,t}$, in USDC
B, B_t, τ	Initial budget, remaining budget, served horizon	$B = \$50$; $B_{t+1} = B_t - c_t$; $\tau \le T$
Reward and wallet pressure		
ν	Failure penalty in utility	$\nu = 0.5$
u_t	Utility: failure-penalized quality used to update the Q-posterior	$u_t = q_t - \nu f_t$
$c_{\max}, \tilde{c}_t$	Cost normalizer and normalized cost	$c_{\max} = \$0.01$; $\tilde{c}_t = \mathrm{clip}(c_t / c_{\max}, 0, 1)$
α, λ_0	Budget-pressure schedule constants	$\alpha = 2.0$, $\lambda_0 = 1$
$\mathrm{burn\_dev}_t$	Normalized burn-rate deviation	Deviation from the linear spend plan $1/T$
$\lambda_t, \lambda_{\mathrm{norm}}$	Raw and bounded wallet-pressure weights	$\lambda_t = \lambda_0 \exp(\alpha \mathrm{burn\_dev}_t)$; $\lambda_{\mathrm{norm}} = \lambda_t / (1 + \lambda_t)$
r_t	Payment-aware reward	$r_t = (1 - \lambda_{\mathrm{norm}}) u_t - \lambda_{\mathrm{norm}} \tilde{c}_t$
PA-DCT posteriors and selection		
$\bar{c}_a$	Listed/spec cost for arm a	Prior mean for the C-posterior
$\mu_{0,q}, \sigma^2_{0,q}, \sigma^2_q$	Q-prior mean, prior variance, and likelihood variance	0.5, 1.0, 0.09
$\sigma^2_{0,c}, \sigma^2_c$	C-prior variance and likelihood variance	10^{-4}, 10^{-6}
$n^{a,k}_t$	Discounted effective count for cell (a, k)	Shared by Q- and C-posteriors
$S^{a,k}_t, S{c_t}^{a,k}$	Discounted utility and cost sums	Updated only for the chosen arm-bucket cell
γ	Discount factor for both Q- and C-posteriors	0.999 in main results
$\theta^{a,k}_q, \Sigma^{a,k}_q$	Q-posterior mean and variance	Normal–Normal posterior over utility
$\theta^{a,k}_c, \Sigma^{a,k}_c$	C-posterior mean and variance	Normal–Normal posterior over realized cost
$\hat{u}^{a,k}, \hat{c}^{a,k}$	Thompson samples from Q and C posteriors	Drawn from the Q- and C-posteriors at decision time
s_a	Decision-time sampled score	Mirrors r_t with sampled utility and sampled normalized cost
Reported evaluation quantities		
$\bar{q}_T$	Full-horizon mean quality	Exhausted rounds count as $q_t = 0$
R^π_T, PA-Reg$_T$	Cumulative PA reward and empirical PA-regret	PA-Reg$_T = R^{\mathrm{Oracle}}_T - R^\pi_T$; reported as PA-gap/$T$
ROI, τ_{adapt}	Quality per dollar and shock-response time	ROI is efficiency only; AdaptT uses trailing-200 ROI thresholds

B Proofs

This appendix gives the formal support for four claims: reward boundedness (Prop. 1), posterior consistency in the stationary no-discount regime (Prop. 2), a stationary well-specified regret reduction (Thm. 1), and a fixed-schedule adaptation rate after an abrupt change (Thm. 2). The regret theorem is intentionally stated for a clean version of the score; §B.7 records the gap to the full empirical setting.

B.1 Shared assumptions for the proofs

We use the notation in Appendix A and the course report. All regret sums below are over served rounds with nonempty affordable set; if a wallet exhausts early, replace T by the served horizon τ. The posterior arguments use the Normal–Normal mean identities

$$\theta^{a,k}_q = \frac{\kappa_q \mu_{0,q} + S^{a,k}_t}{\kappa_q + n^{a,k}_t}, \qquad \theta^{a,k}_c = \frac{\kappa_c \bar{c}_a + S{c_t}^{a,k}}{\kappa_c + n^{a,k}_t}, \tag{1}$$

where $\kappa_q = \sigma^2_q / \sigma^2_{0,q}$ and $\kappa_c = \sigma^2_c / \sigma^2_{0,c}$. We write a single $n^{a,k}_t$ because each selected call supplies one utility and one cost observation. If quality and cost discounts are split, the same formulas apply to the two posteriors with their own effective counts.

The formal claims use the following assumptions.

- **(A1)** *Stationary or piecewise-stationary cells.* Conditional on the chosen arm a and bucket k, and averaging over the exogenous task draw within that bucket, the joint distribution of (q_t, c_t, f_t) is identical within each phase.
- **(A2)** *Sub-Gaussian noise.* The utility u_t given cell (a, k) is σ_u-sub-Gaussian around mean $\mu^{a,k}_u$; the realized cost c_t is $\sigma_{c,\mathrm{true}}$-sub-Gaussian around mean $\mu^{a,k}_c$. The policy uses fixed proxy variances σ^2_q, σ^2_c, possibly different from the true variances.
- **(A3)** *Bounded means.* $|\mu^{a,k}_u| \le 1$ and $\mu^{a,k}_c \in [0, c_{\max}]$ for every (a, k).

The stationary regret reduction further uses two restrictions that are not needed by the boundedness, consistency, or adaptation claims:

- **(A2′)** *Well-specified product model.* The policy's proxy likelihoods are the true likelihoods: utility and cost observations are conditionally independent Gaussians around $(\mu^{a,k}_u, \mu^{a,k}_c)$ with variances $\sigma^2_q = \sigma^2_u$ and $\sigma^2_c = \sigma^2_{c,\mathrm{true}}$, and the prior factorizes as the policy's independent Gaussian Q/C priors.
- **(A3′)** *Clean score.* The regret reduction is stated for the unclipped score $(1 - \lambda_{\mathrm{norm}})\hat{u}^{a,k} - \lambda_{\mathrm{norm}} \hat{c}^{a,k} / c_{\max}$, or equivalently for trajectories on which the implementation's cost-sample clip in the PA-DCT score is inactive.

Propositions 1–2 and Theorem 2 use only (A1)–(A3); only Theorem 1 uses (A2′)–(A3′).

B.2 Proposition 1: reward boundedness

Proposition 1 (Reward boundedness). *For every round t with $0 \le \nu \le 1$, the payment-aware reward satisfies $r_t \in [-1, +1]$.*

Proof. Since $q_t \in [0,1]$, $f_t \in \{0,1\}$, and $0 \le \nu \le 1$, we have $u_t = q_t$ when $f_t = 0$ and $u_t = q_t - \nu \in [-1,1]$ when $f_t = 1$. Thus $|u_t| \le 1$; also $\tilde{c}_t \in [0,1]$ and $\lambda_{\text{norm}} \in (0,1)$. Hence $|r_t| \le (1-\lambda_{\text{norm}})|u_t| + \lambda_{\text{norm}}|\tilde{c}_t| \le (1-\lambda_{\text{norm}}) + \lambda_{\text{norm}} = 1$. □

B.3 Proposition 2: posterior consistency

Proposition 2 (Posterior consistency, stationary regime). *Fix (a,k) and assume the utility distribution at (a,k) is stationary with finite variance and mean $\mu_u^{a,k}$. With no discount ($\gamma = 1$), the Q-posterior mean $\theta_q^{a,k}$ converges almost surely to $\mu_u^{a,k}$ as the cell's pull count $n^{a,k} \to \infty$. The same holds for the C-posterior and $\mu_c^{a,k}$.*

Proof. Let $\{u_t^{(i)}\}_{i=1}^n$ be the utility observations in cell (a,k) after n pulls. By (A1) and the strong law of large numbers, $S_t^{a,k}/n_t^{a,k} = n^{-1}\sum_i u_t^{(i)} \to \mu_u^{a,k}$ almost surely. Dividing the first expression in Eq. (1) by $n_t^{a,k}$ in numerator and denominator gives $\theta_q^{a,k} \to \mu_u^{a,k}$ because $\kappa_q/n_t^{a,k} \to 0$. The cost posterior is identical with $\bar{c}_a, S_{c\,t}^{a,k}, \kappa_c$ in place of $\mu_{0,q}, S_t^{a,k}, \kappa_q$. □

B.4 Lemma: discounted effective sample size

Lemma 1 (Effective sample size). *For any $\gamma \in (0,1)$ and any sequence of pulls of cell (a,k),*

$$n_t^{a,k} \le \frac{1-\gamma^{t+1}}{1-\gamma} \le \frac{1}{1-\gamma}.$$

The first bound is tight when (a,k) is pulled at every round.

Proof. If $\mathcal{T}_t^{a,k} \subseteq \{0,\dots,t\}$ is the set of pull rounds for cell (a,k), the round-based discount rule gives $n_t^{a,k} = \sum_{s\in\mathcal{T}_t^{a,k}} \gamma^{t-s} \le \sum_{j=0}^{t}\gamma^j = (1-\gamma^{t+1})/(1-\gamma)$. □

B.5 Theorem 1: stationary regret reduction

Theorem 1 (Clean-score regret reduction to Gaussian Thompson sampling). *Under (A1), (A2′), (A3′), and $\gamma = 1$, define the* policy-state PA-regret *of clean-score PA-DCT against the wallet-coupled oracle as*

$$\begin{aligned} g_t(a) &:= (1-\lambda_{\text{norm}})\,\mu_u^{a,k_t} - \lambda_{\text{norm}}\,\mu_c^{a,k_t}/c_{\max}, \\ \text{PA-Reg}_T &:= \sum_{t=0}^{T-1}\big(g_t(a_t^\star) - g_t(a_t)\big), \end{aligned} \tag{2}$$

where $a_t^\star \in \arg\max_{a\in\mathcal{A}_t} g_t(a)$ at the policy's wallet pressure λ_{norm}, and a_t is PA-DCT's chosen arm. Conditional on the bucket-visit sequence, clean-score PA-DCT reduces to per-bucket Gaussian Thompson sampling with a time-varying linear context. For a bucket visited T_k times,

$$\mathbb{E}\Big[\text{PA-Reg}_{T_k}^{\text{cell}\,k}\Big] \le C_0\sqrt{K\,T_k\,\log T_k}, \tag{3}$$

for a constant C_0 depending on the prior and likelihood variances and on $c_{\max}$. Aggregating across K_b buckets with $\sum_k T_k = T$ gives

$$\mathbb{E}\big[\text{PA-Reg}_T\big] \le C\sqrt{K\,K_b\,T\log T}. \tag{4}$$

Proof. For a visit to bucket k, write $w_t = \lambda_{\text{norm}}$ and define the observed feature

$$\phi_t = (1-w_t,\ -w_t/c_{\max}), \qquad \theta^{a,k} = (\mu_u^{a,k},\ \mu_c^{a,k}).$$

Then $g_t(a) = \phi_t^\top\theta^{a,k}$. Under (A2′), the independent Q/C Thompson samples used by PA-DCT imply

$$\begin{aligned} &(1-w_t)\hat{u}^{a,k} - w_t\hat{c}^{a,k}/c_{\max} \\ &\quad\sim \mathcal{N}\Big(\phi_t^\top(\theta_q^{a,k},\theta_c^{a,k}),\,(1-w_t)^2\Sigma_q^{a,k} + w_t^2\Sigma_c^{a,k}/c_{\max}^2\Big), \end{aligned}$$

which is exactly the product-Gaussian posterior sample of $\phi_t^\top\theta^{a,k}$. Thus the clean-score policy is Gaussian Thompson sampling on a linear payoff with a two-dimensional, time-varying feature.

The feature ϕ_t, bucket k_t, and affordable set $\mathcal{A}_t$ are observed before sampling. Bucket visits are determined by the exogenous task stream from the course-report setting, so conditioning on a bucket subsequence does not introduce policy-dependent context selection; the wallet pressure and affordable set are predictable state shared by the policy-state comparator in Eq. (2). Therefore the per-round gap is the standard contextual Thompson-sampling regret for at most K arms.

Applying the Bayesian regret bound for Gaussian-prior Thompson sampling to each bucket subsequence gives Eq. (3). Summing over buckets and using Cauchy–Schwarz,

$$\sum_k \sqrt{T_k\log T_k} \le \sqrt{K_b T\log T}.$$

This gives Eq. (4). □

B.6 Theorem 2: adaptation after an abrupt change

Theorem 2 (Adaptation rate under an abrupt change, fixed schedule). *Fix $\gamma \in (0,1)$ and cell (a,k). Suppose the true utility mean shifts between rounds $t^\star - 1$ and $t^\star$, with pre-shock mean μ_{old}, post-shock mean μ_{new}, and $\Delta = |\mu_{\text{old}} - \mu_{\text{new}}|$. Consider any fixed, or reward-noise-independent, post-shock pull schedule $\{s_1 < \cdots < s_m\} \subseteq [t^\star, t]$ for this cell. Define*

$$A := \gamma^{t-t^\star+1} n_{t^\star-1}^{a,k}, \qquad B := \sum_{i=1}^{m}\gamma^{t-s_i}, \qquad \kappa_q := \sigma_q^2/\sigma_{0,q}^2.$$

Then the Bayesian Q-posterior mean satisfies

$$\Big|\mathbb{E}[\theta_{q,t}^{a,k}] - \mu_{\text{new}}\Big| \le \frac{A\Delta + \kappa_q|\mu_{0,q} - \mu_{\text{new}}|}{\kappa_q + A + B}, \tag{5}$$

where the expectation is over observation noise for the fixed schedule.

Remark 1 (Two regimes). The bound (5) separates a prior term from a stale-data term. When $\kappa_q \gg A + B$, the prior controls the bias. When $\kappa_q \ll A + B$, the leading term is $\Delta A/(A+B)$, which is reduced by discounting and by new post-shock pulls.

Corollary 1 (Recovery horizon, data-dominated regime). *Suppose (a,k) is pulled at every post-shock round, so $\tau := t - t^\star + 1 = m$, and assume the data-dominated regime $\kappa_q \ll A + B$. Then*

$$\left|\mathbb{E}[\theta^{a,k}_{q,t}] - \mu_{\text{new}}\right| \lesssim \Delta\gamma^\tau + O\big(\kappa_q/(A+B)\big),$$

and the leading stale-data term is at most δ once

$$\tau \geq \frac{\log(\Delta/\delta)}{\log(1/\gamma)} = O\big(\log(\Delta/\delta)/(1-\gamma)\big).$$

Proof of Theorem 2. For the fixed schedule, conditioning on the pull times does not change the mean of the included observations. From Eq. (1),

$$\theta^{a,k}_{q,t} = \frac{\kappa_q \mu_{0,q} + S^{a,k}_t}{\kappa_q + n^{a,k}_t}.$$

Splitting the discounted mass into pre- and post-shock pulls gives $n^{a,k}_t = A + B$ and $\mathbb{E}[S^{a,k}_t] = A\mu_{\text{old}} + B\mu_{\text{new}}$. Hence

$$\mathbb{E}[\theta^{a,k}_{q,t}] - \mu_{\text{new}} = \frac{\kappa_q(\mu_{0,q} - \mu_{\text{new}}) + A(\mu_{\text{old}} - \mu_{\text{new}})}{\kappa_q + A + B}.$$

Taking absolute values and using $|\mu_{\text{old}} - \mu_{\text{new}}| = \Delta$ gives Eq. (5). □

Proof of Corollary 1. If the cell is pulled at every post-shock round, then $B = \sum_{j=0}^{\tau-1} \gamma^j = (1-\gamma^\tau)/(1-\gamma)$ and $A = \gamma^\tau n^{a,k}_{t^\star-1}$. By Lemma 1, $n^{a,k}_{t^\star-1} \leq 1/(1-\gamma)$, so

$$\frac{A}{A+B} \leq \frac{\gamma^\tau/(1-\gamma)}{\gamma^\tau/(1-\gamma) + (1-\gamma^\tau)/(1-\gamma)} = \gamma^\tau.$$

Substituting this into Eq. (5) yields the displayed bound; solving $\Delta\gamma^\tau \leq \delta$ gives the stated horizon. □

Remark 2 (Cost-posterior analogue). The same calculation applies to realized-cost shocks by replacing $(\mu_{\text{old}}, \mu_{\text{new}}, \mu_{0,q}, \kappa_q)$ with $(\mu^c_{\text{old}}, \mu^c_{\text{new}}, \bar{c}_a, \kappa_c)$ and using the C-posterior statistic under the same locked discount γ. We state the utility version to avoid duplicating notation; this is the algebra used for S3-style price changes once the changed arm is pulled again.

B.7 Scope of the formal analysis

The results above are intended as component-level formal support rather than a complete regret theorem for the full replay system. The stationary regret reduction assumes a well-specified product-Gaussian model, $\gamma = 1$, and the clean score with inactive cost-sample clipping; the implemented clipped scorer coincides with this policy on trajectories where the clip is inactive. The adaptation result is conditional on a fixed post-shock pull schedule, so it explains how discounted posteriors forget stale evidence after a changed arm is revisited, but does not prove an unconditional Thompson-sampling revisit rate or a cumulative piecewise-stationary regret bound. Finally, the theorem's comparator uses the policy's own wallet pressure and affordable set, whereas the empirical *True Oracle* has its own wallet trajectory.

C Detailed Ablation Study

This appendix reports the component ablations summarized briefly in the course report. The ablation study tests whether PA-DCT is a balanced adaptive policy rather than a per-scenario tuning of one scalar. We remove each conceptual ingredient: payment-aware ranking ($-P$), discounting ($-D$), contextual bucketing ($-C$), and Thompson sampling ($-TS$). We also run a price-shock diagnostic, $-C_{\text{post}}$, which keeps wallet pressure but pins cost to the specification scalar $\bar{c}_a$; this isolates whether a learned realized-cost posterior is necessary in S3.

Table 2: Component ablations on 402Pilot-Bench. Gap is PA-gap/T as mean±std over 30 seeds; other metrics are means. $-C_p$ abbreviates $-C_{\text{post}}$. $\bar{q}_T$ counts unserved rounds as zero; † marks mid-run bankruptcy. AdaptT averages finite threshold-hit times; `n/a`: not defined because no shock occurs; `n.r.`: not run for that scenario.

Scenario	Variant	Gap	ROI	$\bar{q}_T$	AdaptT
S1	Full	0.133±0.004	378	0.797	n/a
S1	$-P$	0.855±0.058	111	0.555†	n/a
S1	$-D$	0.109±0.008	412	0.810	n/a
S1	$-C$	0.116±0.003	390	0.812	n/a
S1	$-TS$	0.130±0.046	535	0.740	n/a
S1	$-C_p$	n.r.	n.r.	n.r.	n.r.
S2	Full	0.166±0.006	357	0.761	1467
S2	$-P$	0.878±0.050	103	0.517†	1611
S2	$-D$	0.170±0.012	399	0.745	2249
S2	$-C$	0.158±0.006	368	0.761	1176
S2	$-TS$	0.175±0.048	537	0.687	1125
S2	$-C_p$	n.r.	n.r.	n.r.	n.r.
S3	Full	0.121±0.004	429	0.831	200
S3	$-P$	0.259±0.036	351	0.840	200
S3	$-D$	0.114±0.009	426	0.838	279
S3	$-C$	0.109±0.003	425	0.848	398
S3	$-TS$	0.145±0.025	552	0.742	2232
S3	$-C_p$	0.144±0.005	423	0.797	200

The table assigns a distinct role to each component. Payment-aware ranking is the solvency mechanism: $-P$ overuses expensive arms, exhausts the budget mid-run in S1/S2, and sharply worsens both PA-gap/T and full-horizon quality; its S2 AdaptT is computed on the 14/30 seeds that hit the ROI threshold before bankruptcy. As a consistency check, $-P$ and Contextual DS-TS remove wallet pressure through independent code paths and RNG streams, and their PA-gap/T values differ by at most 0.007 across scenarios. Discounting is the reliability-adaptation mechanism: removing it slows S2 recovery from 1,467 to 2,249 rounds. The learned cost posterior is the price-adaptation mechanism: $-C_{\text{post}}$ keeps post-shock P-premium share near 3.6% instead of Full PA-DCT's 66.1%, raising PA-gap/T from 0.121 to 0.144 and reducing $\bar{q}_T$ from 0.831 to 0.797 even though the ROI-based AdaptT threshold fires at 200 rounds.

Contextual bucketing provides task-specific separation rather than a monotone aggregate win: pooling all contexts can help market-wide shocks and aggregate means, but it slows S3 opportunity capture (398 vs. 200 rounds, with 1/30 seed missing the threshold). Thompson sampling provides exploration stability: greedy posterior means keep PA-gap/T means close to Full in S1/S2 (0.130 vs. 0.133 and 0.175 vs. 0.166) but show about ninefold larger S1 PA-gap/T seed standard deviation (0.046 vs. 0.005) and miss the S3 price shock in 6/30 seeds. Together, these roles cover the failure modes an autonomous buyer faces: budget control, non-stationary response in reliability and price, task heterogeneity, and stable exploration.

D Hyperparameter Sensitivity

This appendix reports the limited sensitivity checks behind the locked configuration used in the course report. We sweep the four exposed selection scalars one at a time on S1 and use the ablation panel for the cross-scenario discount check; posterior variances, T, and B are fixed calibration constants rather than tuned per scenario.

D.1 Targeted S1 sensitivity sweep

Each non-default row in Table 3 changes one scalar while holding the others fixed, using 10 seeds at $T = 10{,}000$. The True Oracle is recomputed for each cell with the same ν and $c_{\max}$ as the tested policy, so PA-GAP/T remains comparable within the sweep.

Table 3: One-at-a-time S1 sensitivity of PA-DCT. Means ± SE over 10 seeds; each non-default row perturbs a single scalar.

Setting	$\bar{q}_T$	Budget %	ROI	PA-GAP/T
DEFAULT	0.796±0.001	42.2±0.3	377.5±2.1	+0.1328±0.0013
$\gamma = 0.99$	0.753±0.001	49.8±0.2	302.4±1.3	+0.2087±0.0013
$\gamma = 0.9995$	0.801±0.002	40.8±0.4	392.6±3.3	+0.1218±0.0016
$\nu = 0.25$	0.795±0.001	42.1±0.4	377.6±3.0	+0.1329±0.0018
$\nu = 1.0$	0.797±0.001	42.1±0.3	378.7±2.3	+0.1324±0.0014
$\alpha = 1.0$	0.788±0.002	36.7±0.3	430.3±3.1	+0.1045±0.0011
$\alpha = 4.0$	0.802±0.001	54.7±0.4	293.5±2.0	+0.1781±0.0016
$c_{\max} = 0.005$	0.787±0.001	41.9±0.4	375.9±3.0	+0.1567±0.0028
$c_{\max} = 0.02$	0.800±0.001	50.5±0.3	316.8±1.9	+0.1490±0.0021

The sweep supports two narrow claims. First, the locked policy is not brittle: no cell bankrupts the wallet, $\bar{q}_T$ stays within $[0.753, 0.802]$, and PA-GAP/T remains in the same order of magnitude. Second, the visible levers are the intended ones: shorter memory worsens stationary performance, and larger α spends more aggressively; ν and $c_{\max}$ have smaller effects in the tested range.

D.2 Discount factor γ

Table 4 gives the cross-scenario endpoint comparison from the $-D$ ablation: full PA-DCT uses $\gamma = 0.999$, while $-D$ sets $\gamma = 1$. The default corresponds to an effective horizon of $1/(1-\gamma) = 10^3$ rounds, chosen to retain S1 evidence while still reacting inside the S2 outage window.

Table 4: Two-point sensitivity to the discount factor γ from the $-D$ ablation. Means over 30 seeds.

Scenario	γ	PA-gap/T	ROI	$\bar{q}_T$	AdaptT
S1	0.999	0.133	378	0.797	n/a
S1	1.000	0.109	412	0.810	n/a
S2	0.999	0.166	357	0.761	1467
S2	1.000	0.170	399	0.745	2249
S3	0.999	0.121	429	0.831	200
S3	1.000	0.114	426	0.838	279

The pattern matches the intended trade-off: no discount is slightly better on stationary S1 endpoints, while discounting improves the shock-response time after S2/S3 changes. This is the empirical counterpart of Cor. 1, where stale evidence decays as γ^{τ} after a changed arm is revisited.

D.3 Other locked scalars

The failure penalty is fixed at $\nu = 0.5$ to give billed timeouts a cost beyond their zero quality score; the S1 sweep shows that doubling or halving it barely moves the reported metrics. The budget-pressure sharpness is fixed at $\alpha = 2$ with $\lambda_0 = 1$, so $\lambda_{\text{norm}} = 0.5$ when spending is on plan and rises smoothly under sustained overshoot; the sweep confirms the expected spend trade-off between $\alpha = 1$ and $\alpha = 4$. We set $c_{\max} = \$0.01$, the highest listed provider price, and the half/double sweep does not change the qualitative conclusion.

The posterior prior and likelihood variances are not swept. The Q-posterior uses a broad mid-range prior with moderate proxy noise, while the C-posterior uses a tight prior/likelihood because realized cost is nearly deterministic except at the S3 price shock. We also do not sweep T and B: the released benchmark fixes $T = 10{,}000$, and $B = \$50$ is calibrated so the premium-only policy hits the budget cliff in the second half of the run.

E x402 Integration Witness

The released artifact includes a local x402 integration witness for the `PaymentExecutor` boundary used by 402Pilot. The witness is not part of 402Pilot-Bench and does not produce any reported measurement. Its purpose is narrower: to show that the selected-arm payment operation can be instantiated as a real HTTP 402 quote–pay–receipt loop. The witness consists of four local components: an Anvil fork carrying USDC contract state, a Python x402 facilitator that verifies and settles exact EVM payments, a FastAPI resource server with x402-protected provider endpoints, and an `X402PaymentExecutor` adapter that maps the HTTP 402 round trip into a 402Pilot `Outcome`.

The witness can be run with `bash scripts/witness_smoke.sh`. A successful run first receives HTTP 402 from an unpaid provider endpoint, then completes one paid call with a positive receipt amount and transaction hash. The live integration test additionally checks that the buyer's local USDC balance on Anvil decreases by the settled amount:

```
X402_INTEGRATION_TEST=1
pytest tests/integration/test_x402_roundtrip.py -v -s
```